

2. Concepts needed to read the study

Sorting, records and correctness

A key is the value used to order data. A record contains that key and associated information, such as a payload or original position. Full sorting returns every record exactly once in order. Stable sorting additionally retains the original order of equal keys. Ranking assigns positions; a full sorter must also realize the output placement. Top-k retrieval returns only a selected subset and is a different task.

For example, records (8,A), (2,B), (8,C) must become (2,B), (8,A), (8,C) under stable ascending sorting. Preserving values alone does not establish that record identities or duplicate order were preserved.

Digital, analog, optical and hybrid

Digital computation represents discrete states, often bits. Analog computation represents numerical quantities through physical signals. Optical describes the physical carrier: an optical system can implement digital Boolean logic or analog arithmetic. Consequently, “optical sorting” does not imply that its comparisons use analog subtraction. The historical bitonic designs and the proposed noisy hybrid adaptation must be distinguished. [4-7]

A hybrid photonic processor divides work between optics and electronics. A typical arithmetic path encodes inputs into light, performs a physical operation, detects the result and performs digital control or further processing. Optical communication alone does not make the arithmetic photonic; the FPGA radix sorter is an important example. [8]

Parallelism, work and dependency

Total work counts all primitive operations. Parallel time counts sequential rounds once dependencies and available resources are fixed. A round can contain many independent operations. Hardware reuse saves simultaneous processing units but may increase rounds. A pipeline overlaps different datasets; a single dataset still traverses its required dependencies.

Fan-in is the number of inputs combined by an operation. Fan-out is the number of destinations receiving an input. Broadcasting to N destinations and summing N inputs are substantial physical requirements even when an abstract model counts each as one stage. The rank sorter makes this distinction particularly visible. [2,7]

Asymptotic notation and physical units

$O(f(N))$ states an asymptotic upper bound; $\Omega(f(N))$ states a lower bound; $\Theta(f(N))$ means matching upper and lower order. Constants and implementation details are suppressed. $\Theta(\log^2 N)$ does not specify nanoseconds, and $\Theta(N^2)$ hardware does not specify square millimetres. All logarithms in the sorting formulas below are base two.

Latency is time until one dataset finishes. Initiation interval or output period is the time between successive datasets in steady operation. Throughput is completed keys or bytes

per second. Power is energy per unit time; energy per sort requires an agreed system boundary and workload duration. These quantities cannot be replaced by one another.

Physical limits that motivate the numerical study

A modulator encodes an electrical input onto an optical signal. A photodetector converts arriving light into an electrical response. A digital-to-analog converter (DAC) produces an analog drive from digital data; an analog-to-digital converter (ADC) samples and quantizes an electrical result. The converter, optical path and electronics can each limit precision. An API receiving floating-point arrays does not reveal the resolution of each internal component. [13-15]

Optical loss reduces useful signal power during propagation, splitting, coupling and component traversal. A fan-out network divides available optical power among destinations unless additional power or amplification is supplied. Longer unregenerated paths can therefore face both weaker signals and accumulated disturbances. Loss is often reported in decibels: a power loss of L dB gives an output/input power ratio of $10^{-(L/10)}$. This relation alone does not specify the resulting numerical error.

Shot noise is associated with discrete photodetection events. Thermal and electronic noise arise in receivers and related circuitry. Laser relative intensity noise describes fluctuations in source intensity. Crosstalk is unwanted coupling between channels. These effects may depend on signal level, bandwidth, shared components and operating conditions; they need not be independent or identically distributed. The current independent Gaussian comparison-error model does not distinguish or calibrate these mechanisms.

Signal-to-noise ratio (SNR) compares useful signal power with noise power under a stated bandwidth and measurement convention. Effective number of bits (ENOB) expresses achieved signal or numerical quality in bit-like terms under a specified test. Nominal converter resolution merely counts representable codes. No SNR-to-ENOB calibration or physical noise budget has been performed for these sorting mappings.

Quantization is a deterministic discretization, while a random-noise model adds statistical perturbations. Either can make nearby keys indistinguishable. Sorting is especially sensitive near equality because the sign of a small difference decides the order. High average arithmetic accuracy does not guarantee a fully correct ordering of many closely spaced keys.

Techniques such as bit slicing or block floating-point representations can alter precision and conversion workload. Bit slicing decomposes a value into smaller bit groups processed separately; block floating point shares a scale or exponent across a group of values. Neither technique is implemented in the current evaluation. Adding either would require a new operation count, representation rule and error analysis rather than treating it as a free precision improvement.

These physical concepts motivate the later calibration and regeneration questions. They are not extra fitted parameters in the present pilot: the executed model contains only the stated quantizers and independent comparison noise. The distinction prevents an illustrative numerical experiment from being presented as a complete device model.

3. The four parameters: T, S, H and D

The comparison uses the five researched sorting structures and counts how their required steps and resources grow with N . It retains all four parameters of $C=(T,S,H,D)$. Device dimensions, clock rates, bit-serial delays, converter costs, optical losses and measured power are excluded from these complexity expressions.

Parameter	What is counted	Simple question
T: time	Dependent processing rounds needed to finish one sort, with the parallel operations provided by the stated structure.	How many rounds must follow one another?
S: space	Peak live logical storage for one input batch, including input, output and necessary intermediate data.	How many values or logical cells must coexist?
H: hardware	Processing-unit positions present at the same time in the stated structure; reused units are counted once.	How many processing positions are provided?
D: optical depth	Longest sequence of optical stages traversed before signal restoration. Electronic decision/storage followed by a fresh optical output ends a segment.	How many optical stages can one signal traverse without restoration?

The counting units remain simple

N is the number of records. S is reported in logical words/cells so that a record, index or rank needs $O(1)$ words. This is the already adopted N -only normalization of the original bit-based S ; it does not change S into area. H counts word-processing positions, not transistors or square millimetres. D counts optical blocks or links, not nanoseconds, metres or individual lenses.

Retain the sorting structure; omit device-specific costs

The distinction between a spatial pipeline and reused processing arrays is part of the researched structure. Keeping that distinction lets H remain useful. Similarly, D uses the optical/electronic boundaries shown in the source architecture. Removing those structural facts as well would make H and D undefined; that is not the purpose of this comparison.

Thus the table is an algorithmic comparison of the five published structures. It is not a claim that every possible implementation of the same sorting algorithm has identical H or D . No additional binary reduction circuit or undocumented photonic cascade is introduced.

3.1 Rules used consistently in the table

Rule	How it is applied
One whole-key operation	A compare-exchange, arithmetic operation or record move has unit cost. Physical key serialization and converter precision are evaluated later.
Parallel operations	Independent operations in a source stage happen concurrently. Louri's original broadcast, wide column sum and parallel output placement remain source-level primitives.
One input batch	S counts live data for one sort. Installed registers for many overlapping pipeline batches and stored copies of the entire schedule are not added to this logical-space count.
Processing units versus memory	H counts processing positions. Memory capacity is recorded by S, so fixed processing resources can coexist with growing S. A whole $N \times N$ array is not counted as one scalar processing unit.
Optical path boundaries	D restarts at data restoration, including digital data decisions followed by fresh optical encoding. Restoring only the comparison control does not restore a passively routed payload.
N-only radix row	Keep key format, digit range and the number of processing streams fixed while increasing N. This is stated beside the row; growing the cluster is a separate scaling question.

Only two source qualifications need special attention

Pipelined optical C&E: the paper explicitly gives passive and restoring exchange variants. T, S and H have the same orders; D differs. Both D values appear in the same table cell rather than silently choosing one. [4, p.1725, Figs.5-6]

Radix: a single-variable result requires a declared resource scaling rule. The row uses a fixed number of processing streams and the paper's fixed key format. It does not choose a particular FPGA clock, board model or optical bandwidth. Without holding the processing resources fixed, T and H would require another variable. [8]

Notation

For bitonic sorting, $m = \log_2 N$, $K = m(m+1)/2$ is the number of active compare-exchange layers, and $P = m^2 - m + 1$ is the researched fixed-shuffle schedule length, including bypass steps. K and P are derived from N. For radix, q is the number of digit passes and R is the digit range; both are constants in its table row.

$O(f(N))$ describes an asymptotic upper bound. The derivations below establish the stated growth orders under these counting rules. These are structural complexity results, not Q.ANT timing or energy predictions. T counts rounds throughout the main table; there is no separate sequential-time column.

4. Historical models and their relationship to C

Thompson: area-time complexity for VLSI (1979)

Thompson studies physical chip area and computation time in a VLSI model, using the discrete Fourier transform as the central problem. The first page of the supplied paper states an area-time-squared lower bound, $AT^2 \geq N^2/16$, within its model. Wire layout and communication matter; the number of active arithmetic elements alone does not determine physical size. [1, p.81]

The connection to this thesis is the trade-off between spatial hardware and execution time. However, Thompson's physical area A is neither logical storage S nor a raw component count H . The DFT bound is not imported as a sorting lower bound. To turn H into area, a separate model for cell dimensions, wires, optical routing, memory and peripheral circuits would be required.

Caulfield: space-time complexity in optical computing (1991)

Caulfield describes normalized spatial, temporal and fan-in resources. Under the paper's operation definitions, the relation $X_A X_T X_F \geq X_P$ expresses that area, time and fan-in together must support the required computational work. X_P denotes the problem-operation measure, not this document's bitonic processing-step symbol P . [2, pp.373-378]

This is directly relevant to rank sorting: many pair comparisons and wide sums can be packed into a small number of abstract stages by allocating many spatial channels and large fan-in. A constant stage count is therefore not free computation. C retains the time/hardware distinction, while fan-in and fan-out remain explicit metadata. No claim is made that the four coordinates alone replace all of Caulfield's assumptions.

Li and Monticone: spatial complexity of optical computing (2025)

This newer work connects physical wave-device dimensions to the structure of mathematical operators, including communication across cuts and overlapping nonlocality. In plain terms, information that must cross a spatial division requires sufficient independent physical channels. Structural sparsity can alter the required device geometry. It is a modern complement to the historical models, not itself an older historical paper. [3]

Its contribution to the present analysis is a warning against equating sparse arithmetic with small total hardware. Bitonic has sparse pairwise operations but must route values between pairings. The whole sorter is nonlinear, so a result for a linear operator cannot be applied unchanged to the complete sorting task. Physical thickness or cross-section is not the same thing as D , the number of unregenerated stages.

Source	Shared insight	Not interchangeable with C
Thompson [1]	Execution time depends on spatial resources and communication.	Physical A $\neq$ logical S; gate count alone does not determine layout area.
Caulfield [2]	Wide parallelism and fan-in can reduce abstract step count.	N-input summation needs its own primitive and precision assumptions.
Li & Monticone [3]	Communication structure constrains physical size.	Device dimensions are not unregenerated optical depth; linear bounds are not full-sort bounds.
Current framework [F1]	Compare T, S, H and D under explicit execution assumptions.	An operational synthesis, not a new universal complexity theorem.

The thesis contribution is the disciplined application of these resource ideas to concrete sorter mappings and reproducible experiments. It does not require claiming the first hardware-aware complexity model or proving a new physical lower bound.

5. Five sorters compared with $C=(T,S,H,D)$

All values use N as the growth variable. T counts dependent rounds, S counts live logical words/cells, H counts processing positions and D counts unregenerated optical stages. Each row is derived separately on the following pages.

Sorter	T	S	H	D
Pipelined bitonic with optical C&E [4]	$O(\log^2 N)$	$O(N)$	$O(N \log^2 N)$	$O(\log^2 N)$ passive; $O(1)$ restoring
Beyette: recirculating bitonic [5]	$O(\log^2 N)$	$O(N)$	$O(N)$	$O(1)$
Desmulliez: perfect-shuffle bitonic [6]	$O(\log^2 N)$	$O(N)$	$O(N)$	$O(1)$
Louri: rank-based broadcast-and-compare [7]	$O(1)^*$	$O(N^2)$	$O(N^2)$	$O(1)$
In-network stable radix [8]	$O(N)^\dagger$	$O(N)$	$O(1)^\dagger$	$O(1)$

* **Rank:** the paper's simultaneous broadcast, comparisons, column sums and output placement are retained. Constant time means a constant number of these parallel stages, with quadratic resources; it does not mean constant total work.

† **Radix:** fixed key format and fixed number of processing streams. The paper uses eight 4-bit passes for 32-bit keys. H counts processing resources, while the memory holding N records is counted separately under S . If the cluster grows with N , this row must be recalculated.

Pipeline D: passive exchange retains the optical data signal across the network; restoring exchange renews it at each stage. The paper describes both. Regeneration of the comparison control alone does not justify $D=O(1)$ for a passive data path.

Communication is included in D: an optical transmission between electronic endpoints is an optical stage. Therefore the FPGA radix row has $D=O(1)$, even though its sorting arithmetic is electronic. If D counted only optical arithmetic, that particular quantity would instead be zero; this table consistently includes optical data transfers.

How to read the comparison

The spatial bitonic pipeline provides all its layers as hardware. Beyette and Desmulliez reuse a linear number of positions. Rank sorting provides a quadratic comparison plane to obtain a constant stage count. The radix row streams the records through a fixed number of processing positions. These are different time/resource trade-offs, not equal-hardware speed measurements.

The next five pages each use the same order: derive T , then S , then H , then D . There is no change to the definitions between sorters.

5.1 Pipelined bitonic sorter: four derivations

Source: Stirik and Athale, Sorting with optical compare-and-exchange modules. The bitonic network is spatially laid out; the paper distinguishes restoring and passive exchange units. [4, pp.1721-1722 and p.1725]

T = O(log²N): count the dependent layers

For $N=2^m$, the merge levels have lengths 2, 4, ..., N. Their respective merge depths are 1, 2, ..., m. All $N/2$ independent compare-exchanges within a layer happen together, but the next layer must receive the previous layer's results.

$$K = 1+2+\dots+m = m(m+1)/2, \text{ with } m=\log_2 N.$$

$$T = O(K) = O(\log^2 N).$$

S = O(N): count the live data for one batch

The current layer contains N records; the next layer replaces them with the same records in a different order. Input/output and temporary values require only a constant number of N-sized collections. A whole-key compare-exchange has constant temporary state. Hence $S=O(N)$ logical words for one batch.

A fully occupied physical pipeline can hold many batches and can have state in every stage. That is installed pipeline storage, not the single-batch logical S used here. Likewise, the paper's spatial module count belongs under H, not S.

H = O(N log²N): count the provided C&E positions

There are $N/2$ word-level C&E positions in each of K spatially provided layers. They remain present even when only one layer is processing a particular batch.

$$H = (N/2)K = N m(m+1)/4 = O(N \log^2 N).$$

D: follow the data through the exchange units

Passive exchange: the payload is routed through K successive exchange layers without data regeneration. Counting a constant number of optical blocks per layer gives $D=O(K)=O(\log^2 N)$. This is the required structural depth, not proof that arbitrarily deep passive cascades remain accurate.

Restoring exchange: the source's regenerative exchange circuit renews the data signal at each layer. The longest segment between renewals contains only a fixed-size stage, so $D=O(1)$. Both cases follow directly from the source's exchange alternatives. [4, Fig.5 versus Fig.6, p.1725]

$$C_{\text{pipeline}} = (O(\log^2 N), O(N), O(N \log^2 N), D),$$

where $D=O(\log^2 N)$ for passive exchange or $O(1)$ for restoring exchange.

5.2 Beyette recirculating sorter: four derivations

Source: Beyette et al., Bitonic sorting using an optoelectronic recirculating architecture. Two optoelectronic processing arrays are reused; digital data are latched and emitted by VCSELs between arrays. [5, Sections 2.A-C]

T = O(log²N): count the repeated processing steps

The fixed-shuffle schedule requires $P=m^2-m+1$ processing steps for $N=2^m$ records. Each step applies a word-level operation to the array in parallel. Treating each such operation as unit cost gives:

$$T = O(P) = O(m^2-m+1) = O(\log^2 N).$$

The original bit-clock expression includes word length and overlap between bit operations. Those details are retained in the historical timing validation. They are not additional variables in this word-level complexity table. [5, Eq.(1) and Section 3.C]

S = O(N): count the records and current state

Each of the two reused arrays holds a batch of N word positions. Their data, current comparison decisions and control state require a constant number of linear collections at word granularity. Thus $S=O(2N+N)=O(N)$. The schedule can be generated as execution proceeds; all P copies of it need not be stored.

H = O(N): count physical positions only once

Each array contains $N/2$ pair-processing positions at word granularity. A fixed number of such arrays provides a total proportional to N . Reusing a position in P rounds does not create P new hardware positions.

$$H = O(2 \times N/2) = O(N).$$

The source constructs these positions from bitwise modules. Counting their bit-level components would reintroduce word length. This table consistently uses word-processing positions for H .

D = O(1): each digital transfer ends the optical segment

The source detects the incoming optical data, processes/latches digital signals and drives new VCSEL output. Its discussion also states that the output can restore losses in the optical path. Thus a signal segment traverses the fixed inter-array optical connection before digital restoration and fresh emission. [5, Section 2.B, p.8167; Section 2.C, p.8169]

Increasing N increases P , the number of times this structure is used. It does not concatenate those separately restored transfers into one unregenerated path. Hence $D=O(1)$, even though $T=O(\log^2 N)$.

$$C_{\text{Beyette}} = (O(\log^2 N), O(N), O(N), O(1)).$$

5.3 Desmulliez perfect-shuffle sorter: four derivations

Source: Desmulliez et al., Perfect-shuffle interconnected bitonic sorter: optoelectronic design. The sorting-node array and memory are reused; data pass through electronic storage and nodes driving fresh optical read beams. [6, Sections 2, 3.C and 4.A-B]

T = O(log²N): count the fixed-shuffle schedule

The first merge uses one processing step. Each of the remaining m-1 merges uses m steps, including bypass steps needed for the fixed interconnection. Therefore:

$$P = 1 + m(m-1) = m^2 - m + 1.$$

$$T = O(P) = O(\log^2 N), \text{ because } m = \log_2 N.$$

The compact network's $K = m(m+1)/2$ active compare-exchange layers and the source's P processing steps are different exact counts, but both have order $\log^2 N$. The source's frequency and bit-serial timing formula are not included here. [6, Section 3, pp.5079-5081]

S = O(N): count data and current masks

The memory array holds N words. Sorting-node state and the current control masks occupy a linear number of logical positions. The source generates successive masks through its regular shuffle procedure, rather than storing a separate complete mask for every step. A constant number of N-sized collections gives $S = O(N)$. [6, Section 3.B-C]

H = O(N): count the reused sorting nodes

The paper explicitly gives $N/2$ sorting nodes for an N-input arrangement. Current control and memory-interface processing add only a constant amount per word position. The node array is reused through all P steps.

$$H = O(N/2 + N) = O(N).$$

D = O(1): use the data-path restoration boundary

The source's shuffle instruction reads a bit plane from memory, sends it through the optical shuffle and sorting nodes, and stores the node output back in memory. Its nodes receive light at detectors, process/store the data electronically and drive output modulators on further read beams. [6, Section 3.C and 4.A-B, pp.5081-5082]

Thus successive loop traversals are separated by digital data storage and renewed optical output. Only a fixed number of optical blocks belongs to one unregenerated segment, giving $D = O(1)$. This conclusion uses the source's data path; the mere existence of an electronic control mask would not by itself establish data regeneration.

$$C_{\text{Desmulliez}} = (O(\log^2 N), O(N), O(N), O(1)).$$

5.4 Rank-based broadcast-and-compare: four derivations

Source: Louri et al., Constant-time parallel sorting algorithm and its optical implementation using smart pixels. Keep the original matrix broadcast, wide column sums and parallel output-selection operations. [7, Section 2 and Section 3]

T = O(1): count the source's fixed sequence of stages

The algorithm broadcasts the inputs, forms differences, resolves ties, thresholds comparisons, sums columns into ranks and places records at their output positions. All $N \times N$ entries in a matrix stage are processed simultaneously. The paper lists seven main steps, including a separate bias-generation step; their number does not grow with N . [7, pp.3088-3090]

$T = O(1) + O(1) + \dots + O(1)$, for a fixed number of stages,
so $T = O(1)$.

Column summation is retained as the paper's wide parallel primitive. Replacing it with a binary tree would change the counting model and produce a different bound. Final output placement is included: this is full sorting, not only rank calculation.

S = O(N²): count the matrix data

The described procedure represents a constant number of $N \times N$ difference, comparison and output-selection matrices. The N inputs, N ranks and N outputs are smaller additional collections. Therefore:

$S = O(N^2 + N) = O(N^2)$.

This is the source matrix procedure's logical space. A sequential program that immediately adds each comparison into a rank counter can use less storage, but it no longer represents all source comparisons simultaneously.

H = O(N²): count the spatial processing positions

The $N \times N$ smart-pixel/comparison plane provides a quadratic number of parallel positions. Output selection requires at most another quadratic plane; rank channels add only $O(N)$. A constant number of planes therefore gives $H = O(N^2)$. The whole plane is not counted as one scalar processor.

D = O(1): the optical sequence does not grow deeper

Increasing N expands the planes and the width of the column sums. It does not add sequential optical stages to the source's fixed broadcast/threshold/rank/output sequence. The maximum optical segment is therefore bounded by a constant, giving $D = O(1)$. Smart-pixel detection and re-emission can divide that fixed sequence further, without changing the asymptotic order. [7, Figs.1-5]

This is a structural depth bound. It does not assert that fan-out power, wide-sum precision or optical dimensions remain independent of N . Those are questions for the physical evaluation.

$C_{\text{rank}} = (O(1), O(N^2), O(N^2), O(1))$.

5.5 In-network stable radix sorter: four derivations

Source and scaling rule: Mizutani et al., In-network stable radix sorter using many FPGAs with high-bandwidth photonics. Increase the record count N for a fixed key format and fixed number of processing streams. The particular board count, frequency and link bandwidth are not needed for the resulting N -only orders. [8]

$T = O(N)$: count the work in the digit passes

A stable counting pass scans N records, forms a histogram over R digit values, computes bucket positions and places the records. Its operation count is $O(N+R)$. There are q digit passes. With a fixed number of processing streams, parallel processing changes constant factors rather than the order in N .

$$T = O(q(N+R)).$$

The source task has $q=8$ passes and $R=16$ digit values:

$$T = O(8(N+16)) = O(N).$$

This abstracts each record-processing/transfer step as unit cost on the fixed structure. The paper's eight passes are not eight scalar time steps: each pass must process the records. [8, Section 2.C-D, Table 1 and Section 3]

$S = O(N)$: count record buffers and counters

Input/output buffers hold $O(N)$ records. The fixed digit range and fixed processing structure require a constant number of histogram/bucket counters and current control values. Hence $S=O(N+R)=O(N)$. Counters up to N occupy one logical word under the stated word convention.

$H = O(1)$: count the processing positions being reused

The number of streams and their counting/routing units is held constant as N grows. More records reuse those positions for longer; they do not require one new processing unit per record. Thus $H=O(1)$. Growing memory capacity is represented by $S=O(N)$, not hidden in a claim of constant total device area.

$D = O(1)$: count one optical link segment

Sorting arithmetic and in-network processing are electronic. Optical transceivers connect electronic FPGA endpoints through the source's communication network. Receiving and electronically forwarding data separates successive optical transmissions. Each optical segment contains a fixed number of link/routing blocks in the stated fixed network, so $D=O(1)$. [8, Section 2.B and Figs.2-3]

This D includes optical communication. It is not a claim of optical radix arithmetic. Increasing the number of transmitted records or digit passes increases time/traffic, not the longest segment between electronic endpoints.

$$C_{\text{radix}} = (O(N), O(N), O(1), O(1))$$

for fixed key format and fixed processing resources.

5.6 Reading the four parameters together

A small bitonic example: $N=16$

Quantity	Value	Parameter it explains
Merge levels	$m=\log_2 16=4$	Used to derive the processing depth.
Active compare-exchange layers	$K=4 \times 5/2=10$	T for the compact bitonic network.
C&E positions in a spatial network	$(16/2) \times 10=80$	H counts all provided layer positions.
C&E positions in one reused layer	$16/2=8$	H stays linear when a fixed number of arrays is reused.
Source fixed-shuffle schedule	$P=4^2-4+1=13$	T uses P for those schedules; P differs from K.
Live records for one batch	16 plus linear temporary state	S is linear, independently of spatial replication.

Why T and D are not the same

Ten dependent layers require ten rounds for that batch. If the data signal is restored after every layer, however, it never traverses ten layers without restoration. T can therefore grow as $\log^2 N$ while D stays constant. A passive payload path instead retains the signal across the layers and has growing D. This distinction is the reason for retaining D in the framework.

Why S and H are not the same

The spatial bitonic network provides all its comparison positions, but a single batch still consists of only N live records plus linear temporary state. Rank sorting explicitly represents quadratic matrix data as well as quadratic processing positions, so both S and H are quadratic there. Memory and processing need separate counts.

What this comparison establishes

The table compares source structures through one set of definitions. It identifies depth, storage and processing-resource trade-offs. It does not identify a universal speed or energy winner: the rows provide different amounts and types of parallel processing. On a fixed modern photonic core, the mappings must account for how those source operations are scheduled onto the available resources.

For context only, sequential execution has $O(N \log^2 N)$ work for bitonic and $O(N^2)$ for pairwise rank sorting. These work counts do not replace T in the main table. The paper's wide rank primitive also does not promise a constant-time operation on an ordinary sequential RAM.

6. Why bitonic versus rank is the focused thesis comparison

The five-architecture review is the breadth of the literature foundation. The experimental core is narrower: compare two algorithmic organizations on a common candidate arithmetic interface. Bitonic creates many dependent rounds but fewer pair evaluations. Rank exposes more independent pair work, followed by rank accumulation and placement. Neither is assumed preferable before resource and correctness constraints are specified.

Bitonic work: $NK/2 = N m(m+1)/4$ pair evaluations.

Pairwise rank work: $N(N-1)/2$ pair evaluations.

For $N=32$, bitonic has 15 active rounds and 240 pair evaluations; rank has 496 independent pair evaluations before summation and placement. Rank's pair phase is wider, but finite device capacity can force several batches. One API call does not establish constant physical time.

Compare-exchange is nonlinear because the destination depends on the ordering of the inputs. A linear MVM can calculate $a-b$, but a decision and routing are still needed. The mathematical identity $r=\max(a-b,0)$, $\min(a,b)=a-r$ and $\max(a,b)=b+r$ motivates a possible nonlinear formulation. It does not establish that a public photonic API implements optical ReLU or arbitrary stage cascading.

The baseline keeps original records electronically and uses candidate accelerator arithmetic for differences. This is deliberately modest and inspectable. The next major scientific extension is native optical nonlinearity and regeneration, if a defensible function and signal path can be modeled. The extension cannot be presented as a Q.ANT capability unless documentation or measurements support it.

PRISM is included as related work because it uses photonic similarity scoring in a selection workflow. Its electronic top-k block selection for a retrieval task is not a full sorting network, and its task-specific efficiency cannot be transferred to returning every record in sorted order. [12]

7. Q.ANT: what its capabilities mean for sorting

The task is to sort records, but the candidate accelerator accepts numerical operations. The mapping must therefore identify which part of sorting can use those operations, which parts remain electronic, and what the public evidence actually establishes. Product facts were rechecked on 1 October 2026; software statements use the pinned source snapshot in [10].

Evidence	Meaning for this study	What still needs checking
LNol/TFLN photonic core [9]	Electrically controlled optical circuits can evaluate arithmetic functions.	The exact internal circuit used for the proposed difference operation.
Linear API: $Y=XW^T$ [10]	A pair difference can be expressed with weights +1 and -1.	Device support and efficient execution for two inputs and one output; accepted ranges, padding and tiling.
Periodic nonlinear API [10]	A separate driver-backed function computes $\text{tcos}(u) \cdot v$.	A calibrated comparator or min/max mapping; direct optical chaining.
Python/C interfaces; PCIe Gen4 x8 [9]	Digital arrays can be exchanged with a host application.	Dispatch, transfers, retained weights and device-resident buffers.
8 GOPS nonlinear throughput; 150 W NPU listing [9]	Vendor operating specifications to record with their workload and boundary.	Pair differences per second and measured power for sorting. These figures do not establish either.

Why a subtraction is the first candidate

- Sorting asks whether a is greater than b. The sign of $d=a-b$ answers that question; the difference is a linear operation.
- The decision “is d positive?” is nonlinear. Sending a difference through an MVM does not also perform that decision or exchange the records.
- The pilot therefore separates arithmetic from control: model the difference, then compare its sign and move the original records digitally. This makes the first precision experiment interpretable.

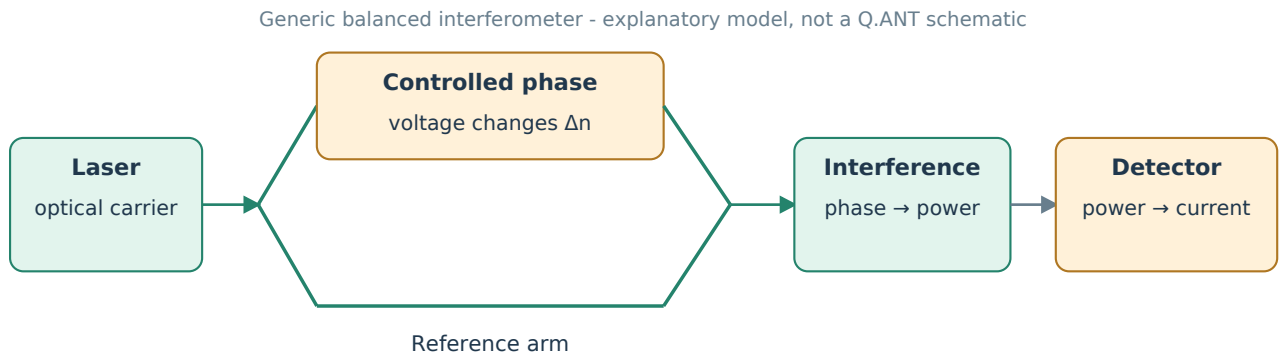
Three kinds of evidence

Documented: public interfaces and vendor specifications. **Calculated:** the sorting equations and logical matrix shapes derived here. **Measured in software:** outcomes of the NumPy pilot. None of these alone is a measurement of the proposed mapping on a Q.ANT device.

Sources: [9] official product page, rechecked 1 October 2026; [10] official SDK at commit 72a2d99f10240b6df3c6d0f636dfa0e2b5d38902. The CPU SDK backend and the NPU have not been run in this evaluation.

7.1 The physics: numbers, light and readout

A digital array does not become optical simply because Python calls a function. Electronic values must control an optical signal; optical transformations then act on that encoding; detection returns an electrical result. The following is a generic explanation of the mechanism, not a reconstruction of Q.ANT's unpublished internal layout.



1. Voltage changes phase

Lithium niobate has an electro-optic response: an applied electric field changes its refractive index. In a waveguide of length L , a change Δn produces the phase change below. λ is the optical wavelength. Over the small-signal Pockels regime, Δn is proportional to the applied electric field; electrode geometry relates that field to voltage.

$$\Delta\phi = (2\pi/\lambda) L \Delta n$$

2. Interference turns phase into an observable signal

Two coherent optical paths recombine. For an ideal balanced, lossless interferometer, one output has $P_{\text{out}} = (P_{\text{in}}/2)[1 + \cos(\Delta\phi)]$. P denotes optical power; $\Delta\phi$ is the relative phase. This periodic response explains how an optical circuit can supply a nonlinear input-output function. It is not automatically a threshold or ReLU, and tcos in the SDK must not be silently replaced by ideal $\cos$.

3. A detector measures power, not a signed real number

In its linear operating regime, a photodiode gives photocurrent $I = \mathcal{R}P$, where $\mathcal{R}$ is responsivity in amperes per watt. Electronics amplify/read this current; digitization produces an array value. Optical power cannot be negative. Signed arithmetic therefore needs an encoding/readout scheme, such as separate positive and negative channels or phase-sensitive detection.

For example, two detected channels can give $I_{\text{diff}} = \mathcal{R}(P_a - P_b)$. This explains physically how a signed difference can be obtained from nonnegative powers. Louri's source explicitly uses two optical channels and electronic subtraction in smart pixels. Which signed-weight mechanism Q.ANT uses for the proposed kernel remains to be confirmed.

Physical background: supplied The Physics of Optical Computing [13]; Louri [7], Section 3.A; and Integrated lithium niobate photonic computing circuit based on electro-optic conversion, Nature Communications 16, 8178 (2025), Fig.1: doi:10.1038/s41467-025-62635-8. The latter is a separate TFLN research circuit, not Q.ANT hardware.